

Quiz 1 Study Guide

The quiz will be in-class on Friday, February 6. Some combination of these problems will be on the quiz.

Problem 1. What does it mean for two events A and B to be *mutually exclusive*?

Problem 2. In order for a function $\mathbb{P}$ to be a *probability measure*, it must satisfy three axioms. State all three axioms.

Problem 3. Let A and B be events, and assume that $\mathbb{P}[A] > 0$. What is the *conditional probability of B given A* ?

Problem 4. What is a k -tuple?

Problem 5. Describe the difference between a combination and a permutation.

Problem 6. A *set* is an _____ collection of elements, all of which are _____.

Problem 7. What does it mean for two events to be *disjoint*?

Problem 8. What is the *sample space* of an experiment? What is an event?

Problem 9. What does it mean for a sequence of events $E_1, E_2, E_3, \dots$ to be *pairwise disjoint*?

Problem 10. Let A and B be events. Define the following terms

- the *complement* of A
- the *union* of A and B
- the *intersection* of A and B

Problem 11. What does it mean for two events A and B to have a positive relationship? A negative relationship?